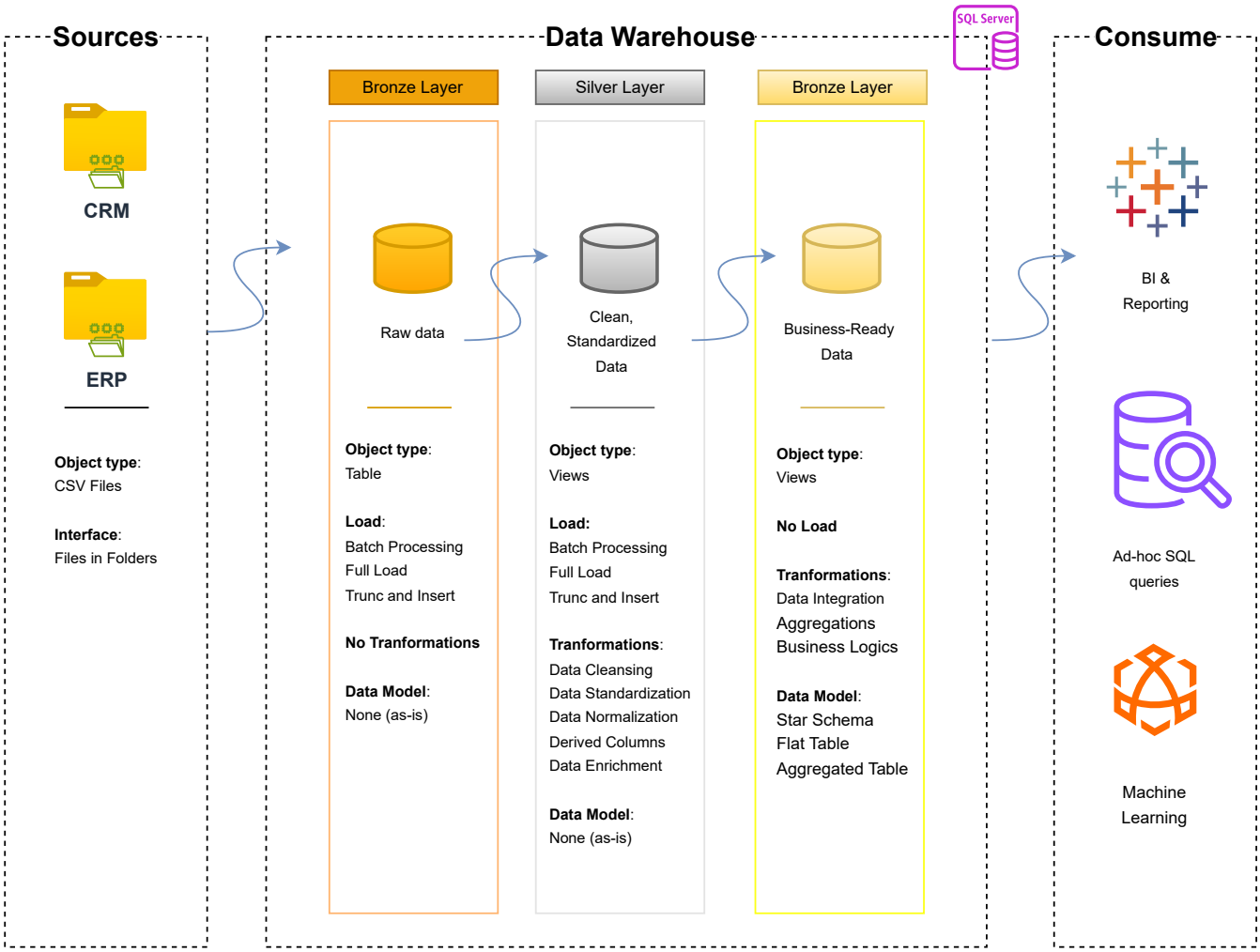High-level Architecture